

Erika Sy

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EDUCATION

University of Wisconsin, Madison <i>Bachelor of Science - Computer Science & Data Science (Honors) Minor: Mathematics & Business</i> Coursework - Introduction to Artificial Intelligence, Honors Computer Graphics, Machine Organization and Programming, Intro to Theory and Probability, Honors Algorithms, Database Management, Consulting	Sep. 2024 - Dec. 2026 Madison, WI.
University of Illinois, Chicago <i>Bachelor of Science - Computer Science</i> Coursework - Software Design, Machine Organization, Data Structures, Programming Practicum, Calc 3, Linear Algebra, Statistics 1, Physics 1	Aug. 2022 - May 2024 Chicago, IL.

SKILLS

Languages: Python, C++, C, Java, JavaScript, TypeScript, HTML, CSS, R, SQL
Frameworks & Libraries: TensorFlow, Keras, PyTorch, FastAPI, React, Next.js, TailwindCSS, scikit-learn, OpenCV, Matplotlib
Tools: Git, Docker, PostgreSQL, Redis, Vercel, Conda, Postman, Figma, Power BI, Unreal Engine, Snap Lens Studio, SolidWorks, Codesys

EXPERIENCE

MCA Program Lead Chief Scientists <i>NASA L'Space</i> - Collaborate with a team to identify and analyze NASA's current challenges, developing innovative solutions to address specific agency pain points in line with the NASA Marshall Space Flight Center's strategic objectives. - Develop and submit a comprehensive proposal responding to a NASA solicitation, enhancing technical writing and project planning skills. - Train as a proposal reviewer, gaining expertise in evaluating the efficacy and viability of project proposals and providing constructive feedback to peers to enhance their submissions. - Acquire practical experience in the end-to-end proposal process, from ideation and writing to review and scoring.	Sep. 2025 - Dec. 2025 Remote
Undergraduate Research Assistant (Dr. Gleicher) <i>University of Wisconsin - Madison</i> - Modernizing and extending a Python backend for interactive visualization and exploration of large text corpora. - Improved interpretability of document embeddings by implementing and evaluating dimensionality-reduction and clustering algorithms. - Identifying salient linguistic and regional patterns across datasets to enhance local and global understanding of corpus structure.	Sep. 2025 - Present Madison, WI.
Business Technology Solutions Intern <i>Abbvie</i> - Identify and implement automation opportunities within the ServiceNow platform to enhance efficiency and reduce costs. - Led 8 employee workshops, used Spreadsheets to document 12 processes, and utilized Visio to map out 5. - Assisted in rebranding the AI landing page, contributing to a more user-friendly interface that improves the customer experience. - Conducted 3 batches of smoke testing using Postman and contributed to the prompt engineering library.	May 2024 - Aug. 2024 Round Lake, IL.
NPWEE Lead SWE <i>NASA L'Space</i> - Collaborate with a team to identify and analyze NASA's current challenges, developing innovative solutions to address specific agency pain points in line with the NASA Marshall Space Flight Center's strategic objectives. - Develop and submit a comprehensive proposal responding to a NASA solicitation, enhancing technical writing and project planning skills. - Train as a proposal reviewer, gaining expertise in evaluating the efficacy and viability of project proposals and providing constructive feedback to peers to enhance their submissions. - Acquire practical experience in the end-to-end proposal process, from ideation and writing to review and scoring.	Sep. 2024 - Present Remote
Undergrad Research Assistant <i>University of Wisconsin - Madison</i> - Engaged in a RoboCup project focusing on advancing autonomous humanoid robots through cutting-edge motion and decision-making technologies. - Contributing significantly to robotic motion analysis using SimRobot to enhance robot walking speed and agility in competitive scenarios.	Sep. 2024 - Present Madison, WI.

Augmented Reality Development Extern

Sep. 2024 - Nov. 2024

Snap Inc.

Remote

- Engineered **5+** AI applications and APIs using **Next.js**, **OpenAI**, **Pinecone**, and **Stripe API**, achieving **98%** accuracy. Managed the project lifecycle from design to development, implementing MVC design patterns to ensure robust and scalable solutions.
- Secured a position among **15,818** selected candidates from **41,803** applicants. Enhanced skills through coaching from Amazon, Bloomberg, and Capital One, focusing on Agile methodologies, CI/CD processes, Git version control, and microservice architecture.

Software Engineering Fellow

May 2024 - Present

HeadStarter

Remote

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Business Technology Solutions Intern

May 2024 - Aug. 2024

Abbvie

Round Lake, IL.

- Identify and implement automation opportunities within the ServiceNow platform to enhance efficiency and reduce costs.
- Led **8** employee workshops, used Spreadsheets to document **12** processes, and utilized Visio to map out **5**.
- Assisted in rebranding the AI landing page, contributing to a more user-friendly interface that improves the customer experience.
- Conducted **3** batches of smoke testing using **Postman** and contributed to the prompt engineering library.

Research Assistant

Feb. 2024 - May 2024

University of Illinois - Chicago

Chicago, IL.

- Contributed to a project to develop a **digital twin** for managing nuclear waste, utilizing **Codesys** and **Unreal Engine**.
- Enhanced the project's efficiency by optimizing the existing simulation code and improving the real-time TCP/IP communication between the physical crane operations and their digital twin representation.
- Examined performance bottlenecks and executed strategic code optimizations to improve the system's efficiency and safety.

Early Research Scholar

Aug. 2023 - May 2024

University of Illinois - Chicago

Chicago, IL.

- Engaged in a forward-looking project to design a whiteboard application using **MediaPipe**, **OpenCV**, **MiniConda**, and **Pycharm**. The application uses hand gesture recognition via the camera, allowing users to intuitively control and interact with a digital whiteboard.
- Participated in solution-oriented team discussions and mentor-led reviews to enhance the application's functionality and user experience.
- Presented research findings and demonstrated the application's capabilities in a comprehensive poster session.

Data Science Workshop Participant

Nov. 2024 - Nov. 2024

University of Illinois - Chicago

Chicago, IL.

- Spent a whole day learning to plot and analyze data using real-life Chicago datasets. Comparing birth rates and other metrics in a mini project, leveraging **Pandas** and **Matplotlib** to tackle social and economic issues through data science.

Seasonal Advisory Intern

May 2023 - May 2023

KPMG

Chicago, IL.

- Investigated innovative technological solutions to help the firm achieve a **net-zero** carbon footprint by 2030.
- Leveraged **Power BI**, **PowerPoint**, and **Excel** to create visually compelling presentations and reports, enhancing stakeholder engagement.
- Wireframe an app using **Figma** that incentivizes sustainable travel choices by rewarding employees with redeemable points for prizes.

PERSONAL PROJECTS

AI Personal Planner | FastAPI, SQLAlchemy, Next.js, React, Tailwind CSS

Oct. 2025 - Present

- Developed a full-stack AI scheduler that integrates Canvas and Google Calendar APIs to synthesize deadlines and generate optimized daily plans.
- Designed a FastAPI backend with an NLP parser and constraint-based optimizer to resolve scheduling conflicts and prioritize tasks.

[Hustle Hub](#) | Python, TypeScript, Next.js, TailwindCSS, FastAPI, PostgreSQL, Redis

Mar. 2025 -

Present

- Making a **FastAPI backend** with **PostgreSQL** and **SQLAlchemy**, adding secure **JWT** login, **form-based auth**, protected **API**

routes, and **token** validation.

- Developed task creation, reactions, and dark-themed UI using **Next.js**, **Zustand**, and **TailwindCSS** for smooth cross-platform **UX**.
- Integrating WebSocket live task feed (via Redis pub/sub), streak leaderboards, push reminders, friend circles, and a mobile app via React Native.

[PlantDocNet](#) | *TensorFlow, Keras, MobileNetV2, scikit-learn, Python*

Apr. 2025 -

Apr. 2025

- Trained a deep learning classifier using **MobileNetV2** on **20 K+** leaf images from a dataset, achieving **89.1%** validation accuracy across 16 disease classes.
- Designed a full **ML pipeline** with image augmentation, data splitting, transfer learning, and early stopping to ensure stable convergence
- Assessed performance using **scikit-learn** metrics (accuracy, precision, recall, confusion matrix) to analyze model robustness and class balance

[Digital Hand Pen Whiteboard](#) | *Python*

Aug. 2023 - Apr. 2024

- Created a gesture recognition whiteboard using **MediaPipe**, **OpenCV**, and **PyCharm** that enabled real-time interactions with a **90%** accuracy rate.
- Participated in solution-oriented team discussions and mentor-led reviews to refine the application's functionality and user experience.
- Showcased the project's innovation and practical applications capabilities at a comprehensive poster session and the **ERSP national conference**.

[Harry Potter Sorting Hat](#) | *CSS, JavaScript, HTML*

Nov. 2023 - Dec. 2023

- Created an interactive **full-stack** game that uses **HTML**, **CSS**, and **JavaScript** to simulate the Sorting Hat's house selection process.
- Developed and implemented an **algorithm** that analyzes and weighs user responses to determine the most suitable house placement.

[Path-Mind](#) | *Arduino, C++, Python, Q-Learning, PyTorch, Matplotlib*

Mar. 2025 - Dec. 2025

- Built a robot with ultrasonic and IR sensors to autonomously solve mazes using wall-following and Q-Learning agents.
- Developed C++ firmware for precise motor control, sensor fusion, and low-latency serial communication with an offboard Python-based AI planner.
- Logged and visualized agent trajectories with Matplotlib; benchmarked heuristics vs. RL, reducing maze completion time by 23%.

[Harry Potter Sorting Hat V2](#) | *Python*

Jul. 2022 - Aug. 2022

- Currently enhancing an interactive full-stack game that uses Python, Flask, CSS, and JavaScript to simulate the Sorting Hat's house selection process.
- Developed and implemented an algorithm that analyzes and weighs user responses to determine the most suitable house placement.

[AI-Powered Support Assistant](#) | *Next.js*

Aug. 2024 - Aug. 2024

- Developed an AI Support Assistant using **Next.js**, **Material-UI**, and **OpenAI API** to enhance user interaction with real-time support.
- Integrated Error Handling and User Experience Features, including loading states, real-time messaging, auto-scrolling, and multi-language support for improved usability.
- Set Up the Development Environment and Deployment Pipeline with **Node.js**, **Vercel**, and comprehensive testing to ensure robust and scalable application performance.

[Space Invaders](#) | *Python*

Dec. 2022 - Dec. 2022

- Developed a Space Invaders clone that utilizes **Python** and **Turtle graphics** to recreate the game with progressive difficulty levels.
- Incorporated a high scoreboard feature to track player achievements, encouraging competition and motivating continuous playability.
- Designed user interface with introductory and "you lose" screens to enhance the gaming experience.

[Harry Potter Sorting Hat](#) | *Python*

Jul. 2022 - Aug. 2022

- Built an interactive **game** to simulate the Sorting Hat's house selection process based on their responses to questions.
- Developed and implemented an algorithm that analyzes and weighs user responses to determine the most suitable house placement.

EXTRACURRICULAR

[Madhacks - Organizer](#)

Jul. 2025 - Oct. 2025

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Cheese Hacks - Hacker

Jul. 2024 - Jul. 2024

UW-Madison Google

Remote

- Developed a real-time object detection tool for visually impaired users, utilizing YOLOv8 for object identification and TTS for verbal feedback.
- Implemented live webcam feed analysis to identify objects and their attributes to enhance user navigation and interaction with their environment.
- Enhanced accessibility and independence for visually impaired individuals by transforming visual data into auditory information.

Madhacks - Hacker

Jul. 2024 - Jul. 2024

MadHacks

Wi. Madison

- Engineered a **full-stack** social-media productivity app during a hackathon that boosts user productivity through social interaction and gamification.
- Enhanced platform interaction by developing a coin system, user authentication, friend functionality, task management, and profile customization.
- Utilized **JavaScript**, **HTML**, **CSS**, and **Bootstrap** to facilitate rapid development and deployment of a scalable social media platform.

Global Hack Week - Hacker

Jul. 2024 - Jul. 2024

Major League Hacking

Remote

- Created and managed all event-related forms, the organization's email, event room bookings, and external organization relationships.

Community Coordinator & Outreach Chair

Jul. 2023 - May 2024

Latinx Organization for Growth in Computing and Academics

Chicago, IL.

- Created and managed all event-related forms, the organization's email, event room bookings, and external organization relationships.
- Responsible for rebranding the organization's merchandise to align with trends and member preferences.
- Helped organize **6** events throughout the year, including UIC's **first-ever** Tech Week.

Newsletter Chair

May 2023 - May 2024

Women in Computer Science

Chicago, IL.

- Crafted **10** bi-weekly newsletters to inform WiCS Members about upcoming programs and workshops to increase engagement.
- Collaborated in bi-weekly meetings, bringing innovative event ideas to cultivate a sense of community.
- Provided support in orchestrating **13** WiCS events, **3** fundraisers, and **1** woman in tech week, assisting with setup and cleanup.

Experience Team Organizer

Sep. 2023 - Feb. 2024

SparkHacks

Chicago, IL.

- Orchestrated a student-run **hackathon** that had **300** participants collaborate, innovate, and compete over **24** hours and **4** prompts.
- Coordinated with judges on the hackathon's objectives, criteria for evaluation, and the support needed to carry out their roles.
- **Co-led** a Mario Kart tournament with **25** participants to foster community engagement and provide a fun break.
- Analyzed feedback, identified areas of improvement, and implemented strategies to strengthen future events.

Flourish Organizer

May 2023 - Sep. 2023

Association for Computing Machinery

Chicago, IL.

- Flourish was an open-source **conference** hosted at UIC. A joint team between UIC ACM, UIC LUG, The Computer Science Department of UIC, and a host of volunteers and alums organized it.
- Collaborated in weekly meetings, bringing innovative event ideas to cultivate a sense of community.
- Emailed **4** people a day to get sponsors for the conference.

Codeathon - Participant

Dec. 2023 - Dec. 2023

Fitch Group

Chicago, IL.

- In this Codeathon, teams will compete to design and improve the user experience for a US-based charity, BigShouldersFund.
- Big Shoulders Fund is an independent charitable organization that supports a network of **72** schools with demonstrated need in the Chicago area.

Member

Mar. 2023 - May. 2023

Engineering for a Sustainable World

Chicago, IL.

- ESW is a student organization at the University of Illinois, Chicago College of Engineering. The group is focused on creating new technology while being environmentally conscious.

- Help ESW build a structure for their **hydroponics** system.

AWARDS

Ideathon - *Participant*

Apr. 2023

Code Your Dreams

Chicago, IL.

- A Hack for Accessibility event with Google Chicago and Deaf Kids Code aimed at making technology more accessible and inclusive.
- Developed a **wireframe** for a website that features a dynamic calendar that employs web-based search functionality to automatically curate and display relevant STEM events, fostering greater engagement and opportunities for underrepresented groups.
- Awarded the **Best Community Outreach Award** for the application.